

Exploring Continuum Robot Agents For Fire Rescuing in Buildings

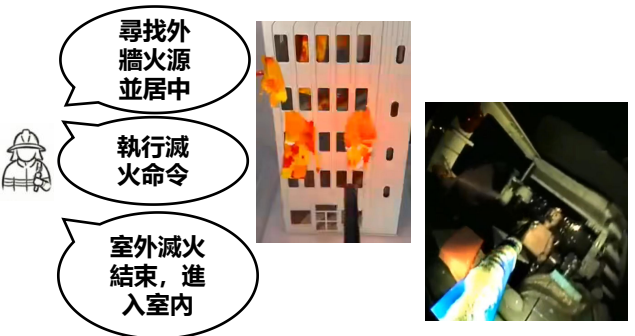
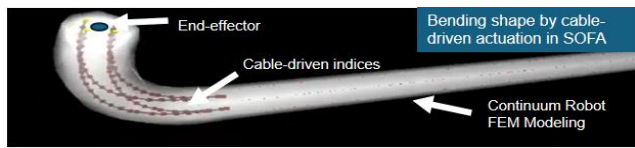
01 Problem & Opportunity

- High-rise, aging, high-density buildings create extreme fire risk.
- Firefighters face poor access, low visibility, and high danger.
- Rigid channels and bulky robots cannot enter narrow blind zones.
- Our solution: Embodied AI + continuum soft robot for rescue.
- Turn fire-fighting infrastructure into an intelligent rescue agent.

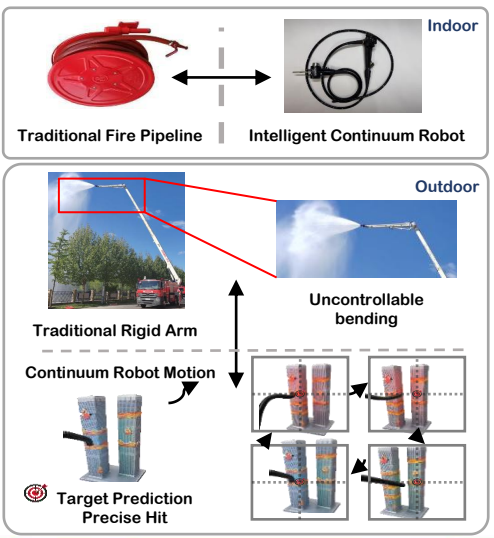
02 System Concept



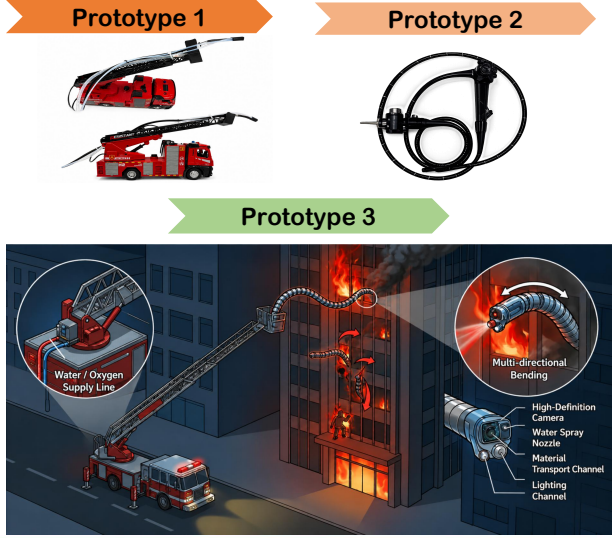
Simulation & Experiment Setup



Experiments



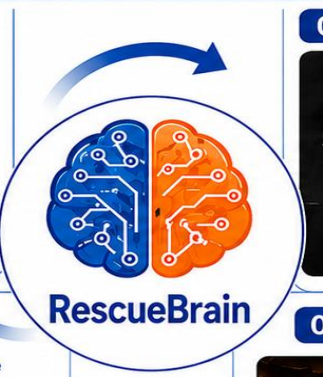
Three Phases Prototype Transition



01 Scene Understanding



03 RescueBrain



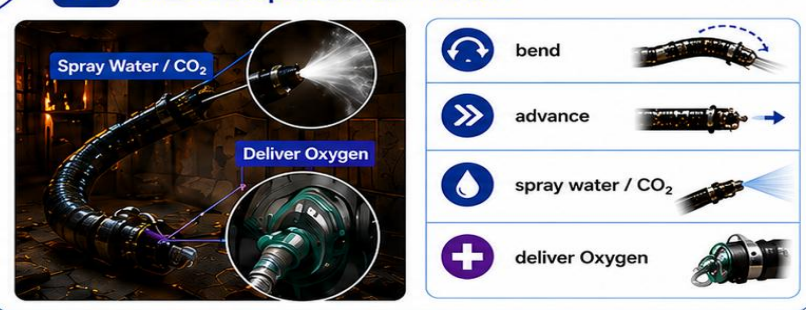
02 Spatial Perception & Affordance



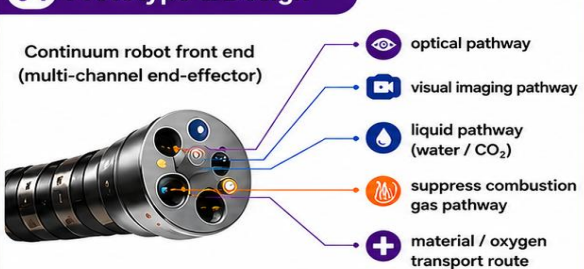
03 Long-term Rescue Planning



04 Closed-loop Continuum Action



04 Prototype & Design



05 Demonstration Scenarios



08 Proven VLA Research Foundation

- Paper accepted
 - EndoVLA, CoRL 2025
 - NeuroVLA, ICRA 2026
 - TrilegVLA, ICRA 2026
- Paper under review
 - EndoVLA-OOV, IEEE-TR 2026
 - BiliVLA, IROS 2026
 - StenosisVLA, IROS 2026

06 Why Our System?

Capability	Traditional Fire Pipe	Rigid Robot	Drone	Our System
Narrow blind-zone access	✗	✗	△	✓
Local fire suppression	✗	△	△	✓
Victim support / oxygen delivery	✗	△	△	✓
Autonomous perception & planning	✗	△	△	✓

07 Key Contributions & Future Work

- Key Contributions
 - Building-integrated continuum rescue concept
 - Embodied AI decision-making with RescueBrain
 - Multi-function rescue: detect, navigate, suppress, deliver
 - Feasibility validated with physical prototype and scenario demos
- Future Work
 - Improve robustness in real smoke and fire environments
 - Strengthen perception, navigation, and autonomy
 - Move toward practical deployment with fire services